

Frequency distribution tables

By size and month

Task 3: Estimate the 90% confidence interval for the same task as the lesson. What changes can you see?

Women shoe sales

US	Germany, GER1												Germany, GER2												Mean	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	GER1	GER2
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00	0.00
4.5	0	0	0	0	1	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0.42	0.08
5	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0.17	0.17
5.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	2	0	1	0	0.08	0.33
6	0	2	0	0	0	0	0	0	0	0	0	0	0	1	3	1	2	0	0	0	0	0	0	0	0.17	0.58
6.5	3	3	1	2	1	0	2	0	2	1	3	4	2	0	2	1	1	2	0	1	2	1	3	0	1.83	1.25
7	0	3	3	4	1	0	1	0	2	0	0	1	0	0	0	4	1	3	1	1	1	3	1	4	1.25	1.58
7.5	1	2	4	1	2	6	4	3	5	8	2	1	2	1	1	3	2	7	9	8	14	8	6	3	3.25	5.33
8	6	10	3	9	1	3	6	8	3	12	3	9	13	6	5	13	5	3	11	6	6	9	8	3	6.08	7.33
8.5	10	10	10	7	14	4	7	7	4	8	7	9	8	5	10	4	5	5	9	7	3	7	9	8	8.08	6.67
9	1	3	8	6	3	1	4	4	0	2	4	2	5	2	2	9	3	1	1	7	2	1	4	2	3.17	3.25
9.5	4	1	2	1	2	2	2	4	5	2	3	2	0	1	1	0	1	2	2	1	7	2	4	2	2.50	1.92
10	0	1	1	1	1	1	3	1	0	0	0	1	0	1	1	0	0	0	2	3	0	2	0	0	0.83	0.75
10.5	1	0	0	0	2	2	4	1	0	3	1	1	0	2	0	0	0	1	0	0	0	0	2	1	1.25	0.50
11.5	0	0	0	2	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	1	0	0	0	0	0.17	0.50
Total	26	35	32	33	28	22	35	28	21	36	25	30	30	19	30	35	20	24	35	38	35	36	37	24		

# populations	Population variance	Samples	Statistic	Variance	Formula
Two	unknown, assumed equal	independent	t	$s_p^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}$	$(\bar{x} - \bar{y}) \pm t_{n_x+n_y-2, \alpha/2} \sqrt{\frac{s_p^2}{n_x} + \frac{s_p^2}{n_y}}$

$$\mathbf{ME} = \textit{reliability factor} * \frac{\textit{standard deviation}}{\sqrt{\textit{sample size}}}$$



$$\mathbf{z_{\alpha/2}} * \frac{\sigma}{\sqrt{n}}$$

$$\mathbf{t_{v, \alpha/2}} * \frac{s}{\sqrt{n}}$$

Sample variance		Pooled variance	Margin of error	90% CI		95% CI	
GER1	GER2						
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.81	0.08	0.45	0.47	-0.13	0.80	-0.23	0.90
0.33	0.33	0.33	0.40	-0.40	0.40	-0.49	0.49
0.08	0.42	0.25	0.35	-0.60	0.10	-0.68	0.18
0.33	0.99	0.66	0.57	-0.99	0.15	-1.11	0.27
1.61	0.93	1.27	0.79	-0.20	1.37	-0.37	1.54
2.02	2.27	2.14	1.02	-1.36	0.69	-1.57	0.91
4.93	16.06	10.50	2.27	-4.35	0.18	-4.83	0.66
12.27	12.24	12.25	2.45	-3.70	1.20	-4.21	1.71
7.72	4.97	6.34	1.76	-0.35	3.18	-0.72	3.55
5.06	6.57	5.81	1.69	-1.77	1.60	-2.13	1.96
1.55	3.72	2.63	1.14	-0.55	1.72	-0.79	1.96
0.70	1.11	0.91	0.67	-0.58	0.75	-0.72	0.89
1.66	0.64	1.15	0.75	0.00	1.50	-0.16	1.66
0.33	2.09	1.21	0.77	-1.10	0.44	-1.27	0.60

	GER1	GER2
n	12	12
t _{90%,df}	1.714	

t 0.05 Lookup>>

At a lower confidence level the intervals are narrower
but obviously with less confidence

1. The two shops are quite similar in sales
2. One shop on size 10.5 out performs the other but the confidence is lower